Engineering Requirements Specification Document

Standard 4K Media Player Operating Software

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V1.2

Sekure Networks A2Z

Revisions

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Sekure Networks	Initial Software Request / Outline	July 31, 2023
1.1	Sekure Networks	Revised software request	Sept 9, 2023
1.2	Sekure Networks	Revised document for clarification	Dec 6, 2023

Review & Approval

Requirements Document Approval History

Approving Party	Version Approved	Signature	Date

Requirements Document Review History

Reviewer	Version Reviewed	Signature	Date

1. Introduction

1.1 Purpose: Create a commercial video file playback software that allows a user to playback audio and video content via files on a micro SD card or a standard USB key. The playback software should be developed with 24/7 playback operation and a software / hardware watchdog to ensure content is continuously playing.

1.2 Scope: Goals of the software development are to achieve a robust and reliable playback of video and audio files.

2. General Description

2.1 Product perspective: Video / Audio playback software made for 24/7 reliability

2.2 Product features: Playback of the standard video and audio with specified formats and containers in the word document AV files and formats.docx

2.3 Operating environment: Indoor retail and professional services environments

2.4 Constraints: Media playback IC hardware, amount of RAM / NAND flash.

2.5 Assumptions and dependencies: AV playback is already supported by the hardware / software platform. We are adding requirements and asking for our own branding / features.

3. System Requirements

3.1 Functional requirements

A: Boot screen required. Anything displayed via the HDMI output before the boot screen should just be a black image (blank screen – nothing displayed) that covers the entire video framebuffer.

B: Boot screen should display the following information in text in the lower third of the image. <Device name> (variable set by user), <SW Version> (set by our company), <Mac Address>, read into memory from HW device. <IP Address> (set by ethernet stack). Boot screen should also display an image centered vertically and horizontally within the first 2/3rds of the image. Please provide the image frame size and our company will provide a graphic to be displayed.

C: Video / Audio playback software should be able to decode and playback AV to be displayed via the HDMI output. Please reference AV files and formats.docx document provided.

D: HDMI output settings should be automatically detected by the software system via EDID. If EDID does not report back the default display should follow CEA 1080p 60Hz.

E: Following the boot screen the player should read the SD card with the goal of finding compatible files and playing them via HDMI. We need to know any constraints or limitations in terms of the file names. Files should be played back in alphabetical order.

Files should be played consecutively and continuously. IE if there are 5 files on the card they should be played in the following order.

FileA
FileB
FileC
FileD
FileE
FileA(back to first file)
FileB
ETC (continuously not stopping playback)

F: If any of the files on the SD card are not able to be decoded and played successfully, they should be skipped and the next available file should be played. Playback should always continue we want to avoid players needing to be physically reset by a user. The system should be autonomous.

G: If there are no files on the SD card that can be decoded and played successfully then there should be a message displayed on screen via HDMI "Please insert a compatible micro-SD card, please check the files on the card."

H: Ethernet addressing should work as DHCP only. No static addressing for now. This is done to simplify the software at this stage. However, if the current software has the ability to do static addressing, then we can accept that method as well.

I: The software should play the videos back-to-back with no black space between videos. If this is not achievable with the hardware / software without significant effort, then please minimize the black space between videos as much as possible. *Note we have tested the sample and the back-to-back video playback is acceptable.

J: Player software needs to be commercial grade and playback reliably 24/7. A watchdog timer would be required to monitor critical processes (in terms of video playback). The watchdog timer should provide a system reset if the video playback stops for any reason.

K: The micro-SD card should also be able to read settings stored in a file called config.txt. These variables in the file are simple configuration settings. IE the customer wants to mute the volume during playback. Wants to set volume to a specific level. Wants to force a resolution. We will provide a complete set of variables and values for them soon. Simple Example below.

Inside the config.txt file is a simple example of a configuration setting.

Hdmivolume=4 (1 – 10 scale)

L: Need a RTC (real time clock) on board to be able to set the player to start playing and end playing during specified hours. This information for the start and end of playing will be configured via the config.txt file. These settings should put the player in low power mode when set to stop (end) playback. The player should end in low power mode and go into standard operational mode when set to start playing.

Config.txt example

openhour=13 (1 pm)
closehour=19 (7 pm)

M: No branding other than our company's branding should be displayed at any time during the operation of the software and video output.

N: Analog audio via the 3.5mm jack will be configurable via the config.txt file

4.External Interface Requirements

4.1 User Interfaces

Users can interface with the media player through specialized PH4 connectors that attach LED pushbuttons to the player. These pushbuttons will have various uses like starting or pausing video playback and changing the volume levels.

Also there will be a multifunctional reset switch on the board for resetting the board.

4.2 Software Interfaces

Not for the initial software release but we would like to discuss and determine the feasibility of developing a web server-based software for the user to setup the player instead of the config.txt file on the micro SD card or USB key. This is just for informational purposes at this time and does NOT need to be developed right now. This interface is not part of the current scope for the initial software request.

5.Configuration Parameters (config.txt)

5.1 Configuration Parameters

These parameters will be stored on the Micro SD card in a simple text file called config.txt

hdmioutput=X (1 - CEA 1080p 60Hz, 2 – CEA 3840 x 2160p 60Hz) more resolutions will be provided after initial software acceptance.
hdmivolume=X (0 – 10) 0 = Mute 10= Full volume
analogvolume=X (0 – 10) 0 = Mute 10 = Full volume
openhour=X (1 – 24 = 24-hour clock) automatic turn on and playback starts
closehour=X (1 – 24 = 24-hour clock) automatic shutdown and playback ends – sleep mode
audioout=X (1 = hdmi (default), 2 = 3.5mm analog, 3 = hdmi and 3.5mm analog simultaneous)
playback=X (1 = continuous play (loop – default) 2 = Attractor video with button playback. More playback modes may be provided after initial software acceptance.

6. Hardware and Operating Specifications

6.1 Background Info

Sekure Networks is requesting a quotation for designing a media player that is capable of decoding and displaying 4K video files via HDMI to a TV / monitor. For reference we have provided an image of a previous media player we designed, this product uses an Allwinner A20 IC. Currently we are considering using Rockchip 3566 in the design but are open to suggestions from the team. The team must be able to provide the hardware design and the operational software (firmware). The player is going to be used indoors in a retail or professional office environment.

6.2 Video & Audio Playback Support

Single 4K60p decoding.

Supported Video Containers for playback: .ts, .mov, .mp4, .mkv (H265/H264)
Supported Audio formats for playback: MP2, MP3, AAC, and WAV

6.3 Peripherals

USB High Speed Host Port – USB A at least one

- USB will be used for connecting USB keys to flash new fw to the device. Also will be able to read and playback video files from this external storage device

HDMI 2.0 Output

Ethernet (1000 BaseT)

3.5mm audio output

PH4 SMT Connectors for connecting pushbuttons equivalent to JST part number S4B-PH-SM4-TB(LF)(SN)

Nand Flash chip or eMMC onboard for bootup and instruction code – Need ability to update the flash via the USB input.

Supercap or other for RTC operation.

- This is for real time clock features so that we can program the player with playback times. Meaning we can say start playing at 9am and stop playing at 9pm (or any configuration of timed playback)

Status LEDs

- Status LED – green indicates that power is on, green flashing indicates fw is being updated, red means content issue (unable to playback current content)

HW Reset button.

- Small switch for hard resetting the power to the device.

MIPI video connector for connecting small LCD panels IE 7” / 8” / 10” / 15” inch etc.

6.4 Other

The player will playback video files based on alphabetic order by default. When playing multiple files, we want to have the files playback with minimal time between file playback (which we call blackspace). Ideally it should play the files back-to-back with no blackspace in-between showing the files.

Hardware watchdog to reset the power on the player if required (player outside of normal operating procedure). ICs usually have a hardware watchdog, and we just want it implemented if available.

Software watchdog should also be implemented tied into several points to monitor the playback of the device and trip and reset if necessary.

Hardware should not require any active cooling fans. But a passive heatsink can be utilized.

Players are required to play video files continuously 24 / 7 operation. The goal is to have a player that will playback 24 / 7 reliably and if there is an issue during playback the system would either trip the hardware or software watchdog resetting the power and resuming playback after bootup.

Operating Temperature range (0C – ~80C)